

Automated End-to-End Cloud Data Pipeline & Analytics Solution

R4-Microsoft Data Engineer ·
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Agenda

01

Problem Statement & Objectives

02

Architecture & Technology Stack

03

ETL Workflow & Azure Blob Storage

04

Snowflake Warehouse & Data Loading

05

Airflow Pipeline & Docker Environment

06

Star Schema, Power BI Dashboard & Results

07

Future Enhancements & Conclusion

Project Introduction

Project Overview

Developed an end-to-end cloud data pipeline that automates data collection, transformation, storage, and analytics within a centralized platform.

Core Objective

Replace manual spreadsheet-based workflows with a scalable and reliable data warehouse that provides a single source of truth.

Business Value

Enable faster reporting, improve data accuracy, and support informed decision-making through interactive Power BI dashboards.

Expected Outcome

Deliver an automated, secure, and scalable analytics solution capable of handling multi-source business data efficiently.

Problem Statement & Project Objectives

The Problem

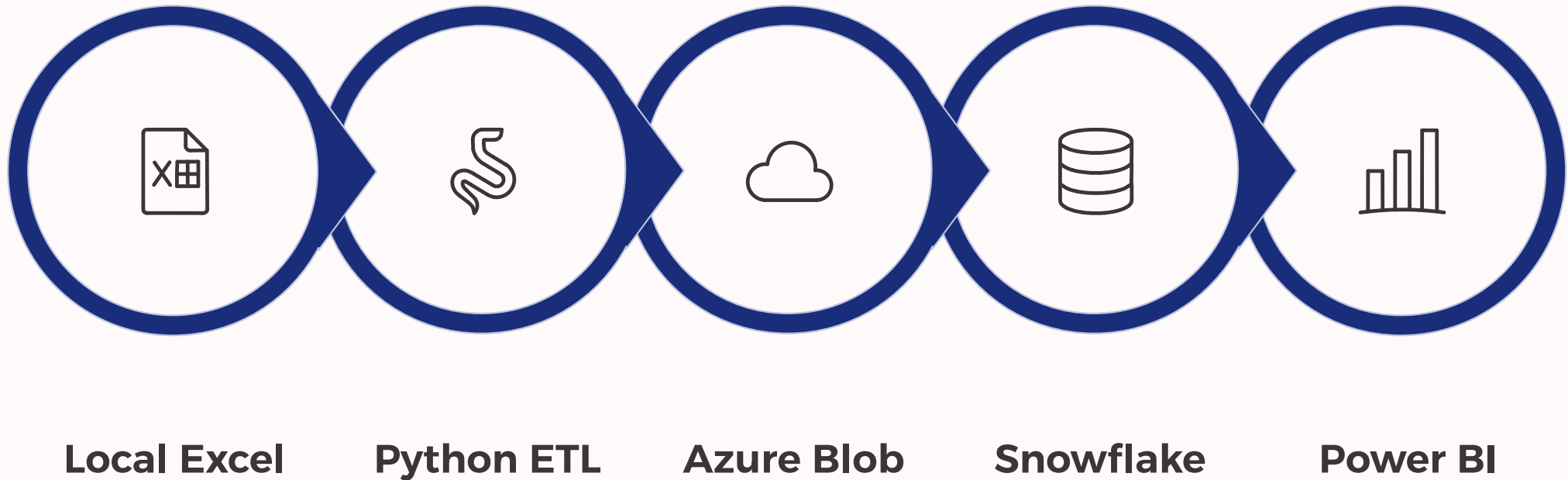
- Data siloed on individual machines — no single source of truth
- No concurrency: Excel files cannot be queried by multiple consumers
- No auditability: manual operations leave no change record
- No refresh automation: every update requires manual rework

Our Objectives

- Automate ingestion & cleaning of 10-entity Excel dataset via Python
- Land data in Azure Blob Storage as a governed raw layer
- Load into Snowflake via COPY INTO with typed relational tables
- Orchestrate all stages as a scheduled Airflow DAG (@daily)
- Expose six Power BI report pages for self-service analytics



Project Architecture



The pipeline moves data through five distinct layers — ingestion, raw cloud storage, warehousing, orchestration, and presentation — each with a single clear responsibility and a single handoff artifact to the next layer.

Technology Stack



Python

pandas + openpyxl for Excel ingestion, cleaning, and CSV export



Snowflake

Cloud data warehouse with decoupled storage & compute



Apache Airflow

DAG scheduling, retries, and task-level observability



Azure Blob Storage

Immutable raw landing zone — raw-gym-data container



Docker Compose

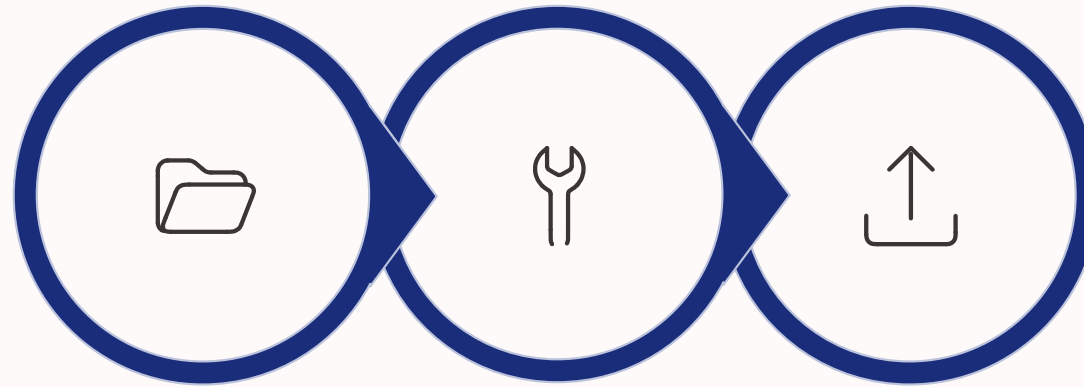
Multi-service local deployment: Airflow + PostgreSQL



Power BI

Six-page analytics dashboard connected to Snowflake

ETL Workflow: Extract → Transform → Load



Extract

Transform

Load

Key Transformations

- Drop **Unnamed:** columns from Excel artifacts
- Strip whitespace, fill nulls with 0 or 'Unknown'
- Parse dates to ISO 8601; clamp negative prices to 0

Code Snippet

```
def clean_file(file_name, df):  
    df.drop_duplicates(inplace=True)  
    for col in df.select_dtypes(['object']):  
        df[col] = df[col].str.strip()  
    df.fillna(0, inplace=True)  
    return df
```

Azure Blob Storage & Snowflake Warehouse

Azure Blob Storage

Single storage account `gymstorage2030` hosts container `raw-gym-data` — the immutable raw landing zone for all 10 cleaned CSV files. Snowflake accesses it via a scoped SAS token (read-only, time-bounded).

```
CREATE OR REPLACE STAGE GYM_DB.PUBLIC.AZURE_STAGE
  URL =
  'azure://gymstorage2030.blob.core.windows.net/raw-gym-
  data'
  CREDENTIALS = (AZURE_SAS_TOKEN = '***');
```

Snowflake Setup

Database `GYM_DB`, schema `GYM_SCHEMA`, and virtual warehouse `COMPUTE_WH` are provisioned. Auto-suspend after 5 minutes eliminates idle compute costs.

```
ALTER WAREHOUSE COMPUTE_WH SET
  AUTO_SUSPEND = 300
  AUTO_RESUME = TRUE;
```


Creating Tables & Loading Data

Target Tables (10 Entities)

- Customers, Employees, Geography, Region
- Product, ProductSubcategory, ProductCostHistory
- SalesHeader, SalesDetails, SalesReturns

```
CREATE TABLE GYM_DB.PUBLIC.Employees (  
  EmployeeKey INT,  
  EmployeeName VARCHAR(100),  
  JobTitle VARCHAR(100),  
  RegionKey INT,  
  HireDate DATE,  
  BaseSalary DECIMAL(10,2),  
  CommissionRate DECIMAL(5,2)  
);
```

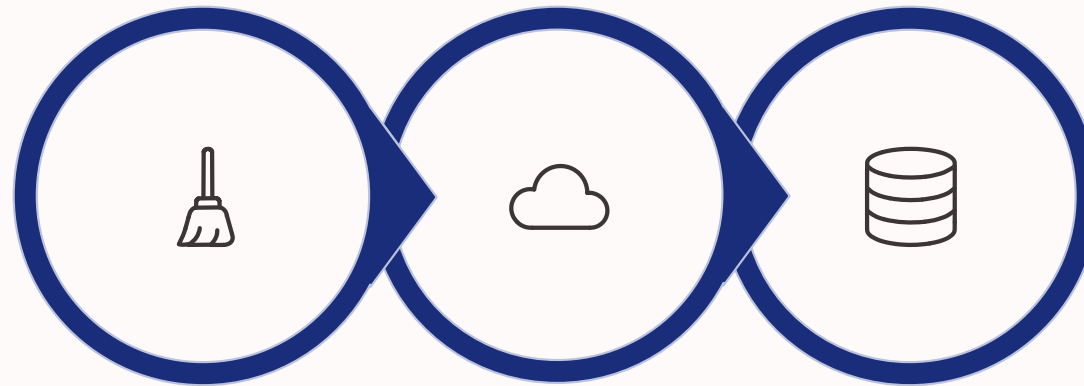
Bulk Load via COPY INTO

All 10 CSV files are loaded using Snowflake's bulk loader — no row-by-row INSERT logic required.

```
COPY INTO GYM_DB.PUBLIC.Employees  
FROM @GYM_DB.PUBLIC.AZURE_STAGE/Employees.csv  
FILE_FORMAT = (TYPE='CSV'  
  FIELD_DELIMITER=','  
  SKIP_HEADER=1);
```

- ❏ All 10 tables follow the identical COPY INTO pattern, differing only in target table name and source filename.

Airflow Pipeline & Docker Environment



Clean Files

Upload CSV

**Copy to
Snowflake**

DAG `gym_complete_data_pipeline` runs daily with linear task dependencies — clean → upload → load — completing in ~18 seconds per run.

Docker Compose Topology

webserver-1

Airflow UI on port 8082

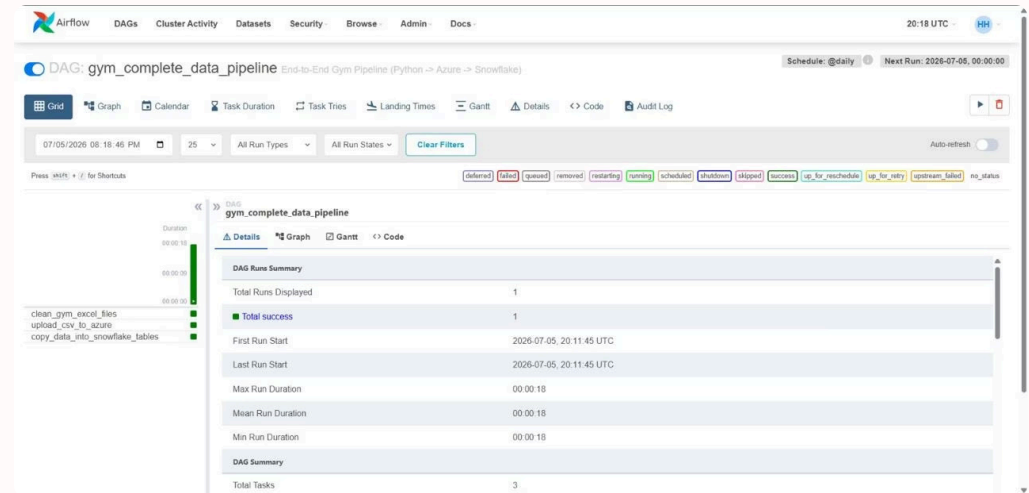
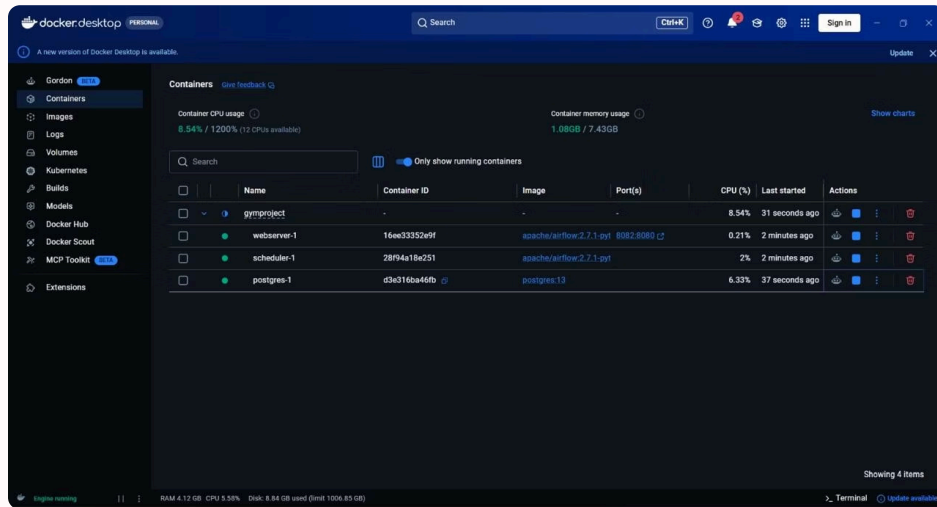
scheduler-1

Parses DAGs, triggers runs

postgres-1

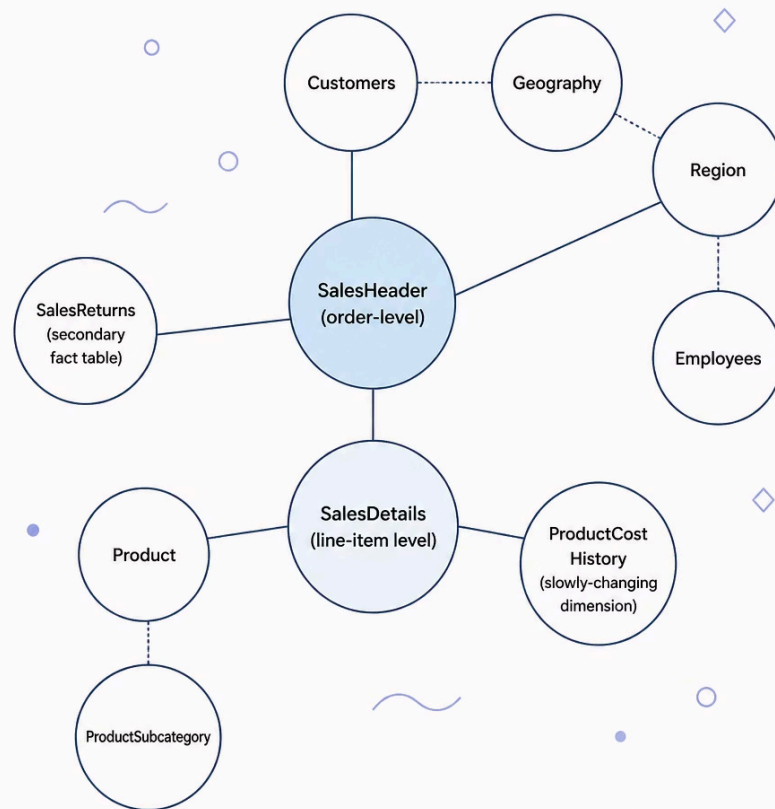
Metadata DB (postgres:13)

Pipeline Orchestration via Apache Airflow



Star Schema Design & Power BI Dashboard

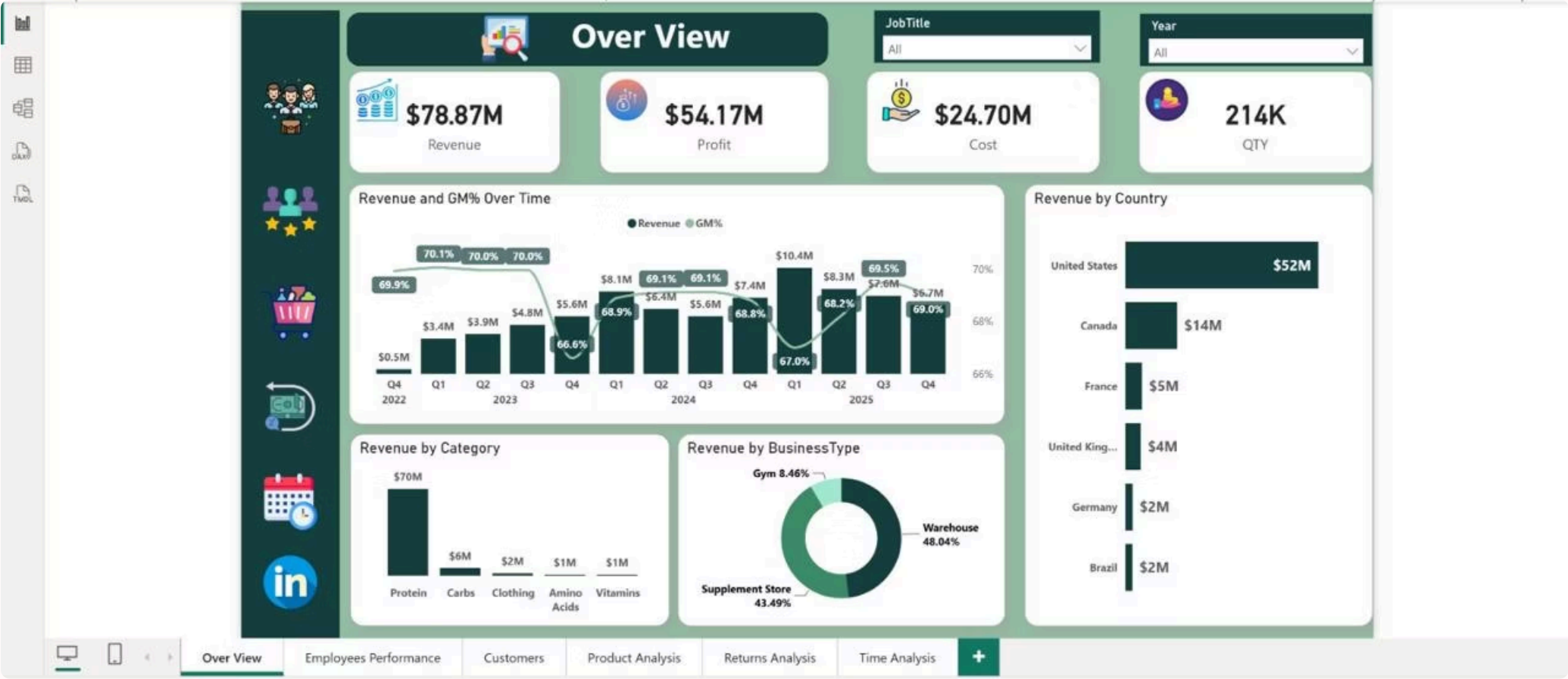
Retail Sales Star Schema



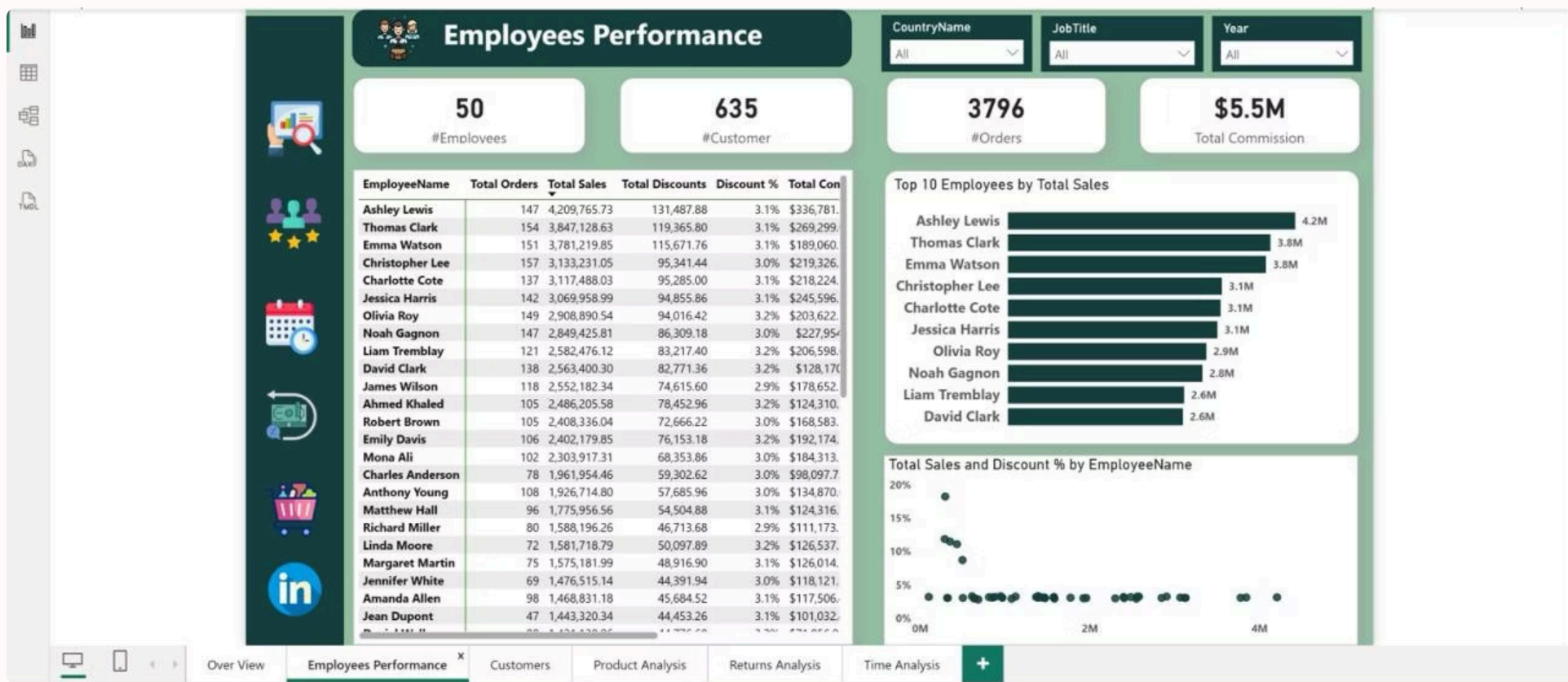
Power BI – Six Report Pages

- **Over View:** Revenue \$78.87M, Profit \$54.17M, GM% trend Q4 2022–Q4 2025
- **Employees Performance:** 50 employees, \$5.5M total commission, top-10 sales ranking
- **Customers:** 635 customers, \$20.78K AOV, churn & dormant account analysis
- **Product Analysis:** GM% 68.7%, returns 8.1%, top products by profit & margin
- **Returns Analysis:** \$6.39M returns, seasonal trends, subcategory breakdown
- **Time Analysis:** YTD revenue, monthly seasonality, prior-year variance

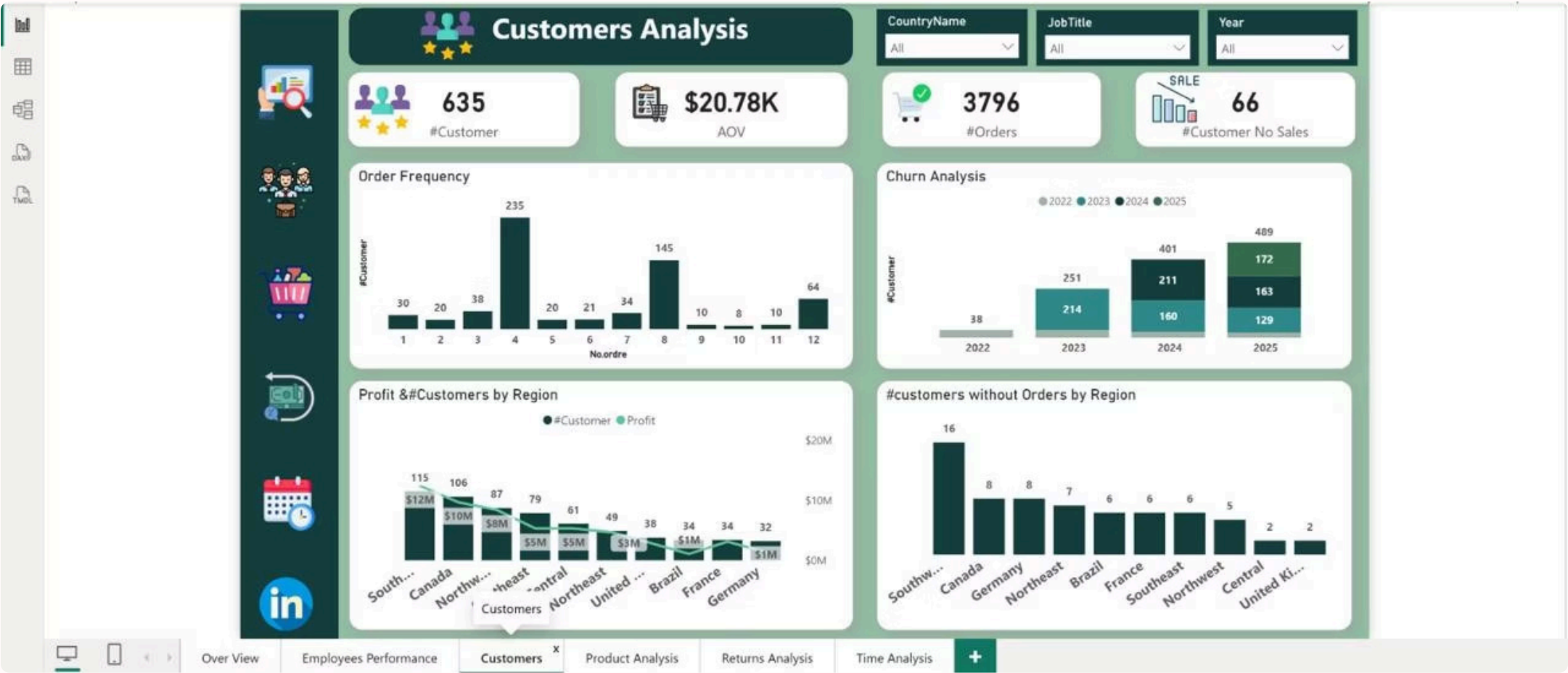
Executive Performance Overview



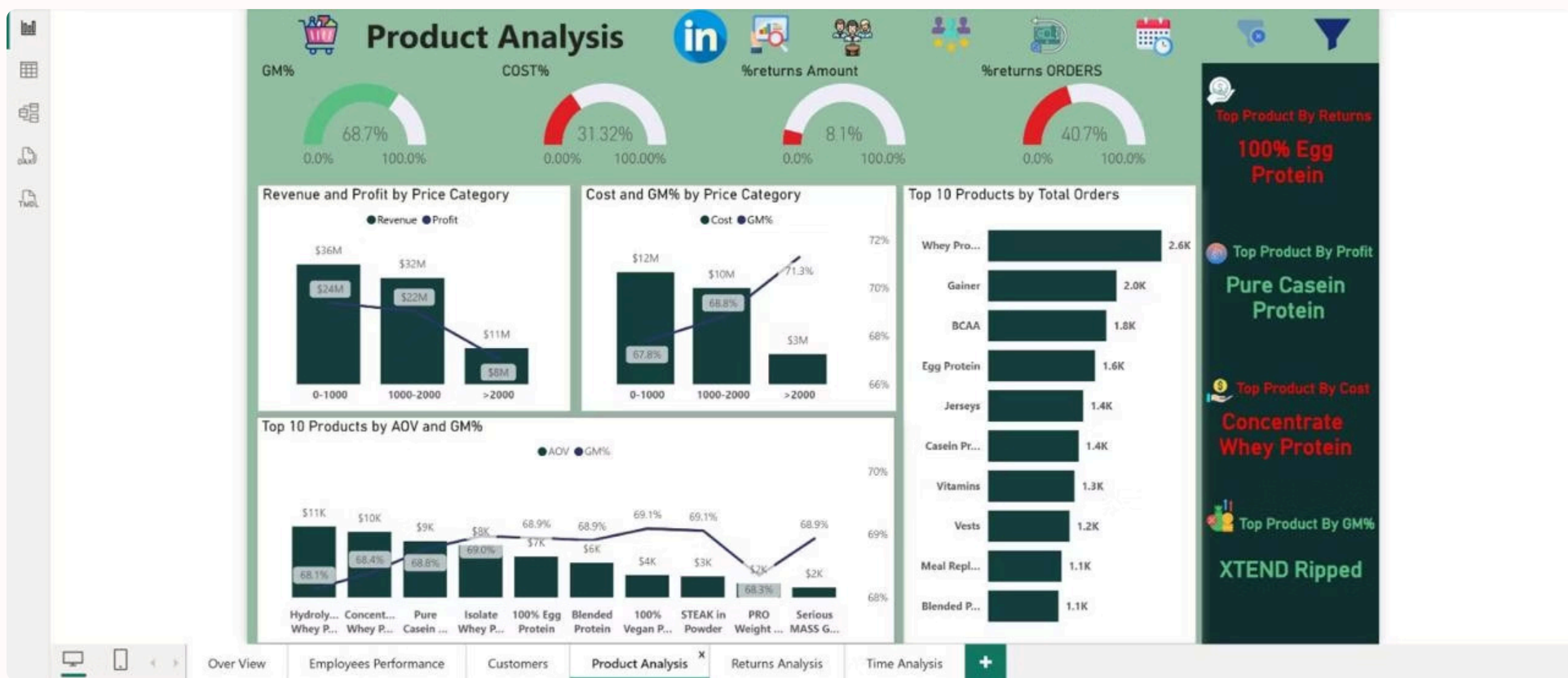
Workforce & Sales Efficiency



Executive Performance Overview: Revenue & Market Trends



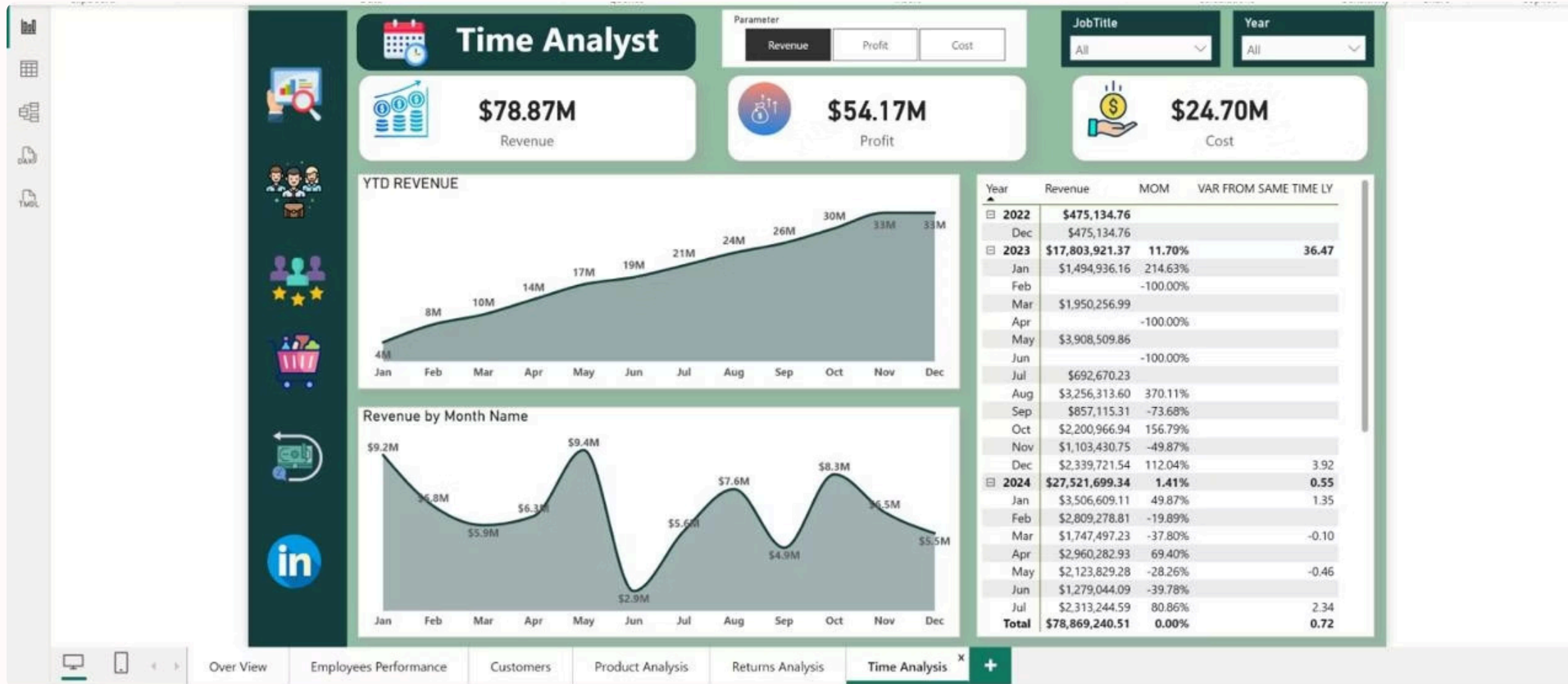
Workforce Efficiency & Sales Contribution



Operational Bottlenecks vs. Strategic Architecture



End-to-End Modern Data Stack (MDS) Architecture



Project Conclusion & Value Delivered

Successful End-to-End Pipeline

Designed and implemented a fully automated cloud data pipeline that seamlessly transfers data from local sources to Azure Blob Storage, Snowflake, and Power BI using scheduled ETL workflows.

Data-Driven Decision Making

Transformed raw, disconnected datasets into interactive Power BI dashboards, enabling faster reporting, KPI monitoring, and informed business decisions.

Scalable & Future-Ready Architecture

Built a modular architecture using Docker, Snowflake, and Apache Airflow, allowing the solution to scale efficiently and support future business growth.

Questions?

Thank you for your attention to this Cloud Engineering showcase.

Automated Pipeline | Power BI Intelligence | Snowflake Warehousing

Digital Egypt Pioneers 2025ons & Discussion